

1.processid

```
sushmitha@DESKTOP-607RQ7V:~$ nano myprocessid.c
sushmitha@DESKTOP-607RQ7V:~$ gcc myprocessid.c -o myprocessid
sushmitha@DESKTOP-607RQ7V:~$ ./myprocessid
My Process ID (PID) = 718
sushmitha@DESKTOP-607RQ7V:~$
```

```
sushmitha@DESKTOP-607RQ7V: ~
GNU nano 8.7.1 mypro
#include <stdio.h>
#include <unistd.h>
int main()
{
printf("My Process ID (PID) = %d\n", getpid());
return 0;
}
```

2.creating child using fork()

```
sushmitha@DESKTOP-607RQ7V:~$ nano childusingfork.c
sushmitha@DESKTOP-607RQ7V:~$ gcc childusingfork.c -o childusingfork
sushmitha@DESKTOP-607RQ7V:~$ ./childusingfork
Parent Process
PID = 916
Child Process
PID = 917
sushmitha@DESKTOP-607RQ7V:~$
```

```
sushmitha@DESKTOP-6O7RQ7V: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
int main()  
{  
    pid_t pid = fork();  
    if(pid == 0)  
    {  
        printf("Child Process\n");  
        printf("PID = %d\n", getpid());  
    }  
    else  
    {  
        printf("Parent Process\n");  
        printf("PID = %d\n", getpid());  
    }  
    return 0;  
}
```

3. Parent Waits for child(wait())

```
sushmitha@DESKTOP-6O7RQ7V:~$ nano parentwaitsforchild.c  
sushmitha@DESKTOP-6O7RQ7V:~$ gcc parentwaitsforchild.c -o parentwaitsforchild  
sushmitha@DESKTOP-6O7RQ7V:~$ .\parentwaitsforchild  
.parentwaitsforchild: command not found  
sushmitha@DESKTOP-6O7RQ7V:~$
```

```
sushmitha@DESKTOP-607RQ7V: ~  
GNU nano 8.7.1 parentwa:  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
int main()  
{  
pid_t pid = fork();  
if(pid == 0)  
{  
printf("Child Process Executing...\n");  
sleep(3);  
printf("Child Finished\n");  
}  
else  
{  
wait(NULL);  
printf("Parent Process Continues\n");  
}  
return 0;  
}
```

4. Multiple Child Processes

```
sushmitha@DESKTOP-607RQ7V:~$ nano multiplechid.c  
sushmitha@DESKTOP-607RQ7V:~$ gcc multiplechid.c -o multiplechid  
sushmitha@DESKTOP-607RQ7V:~$ .\multiplechid  
.multiplechid: command not found  
sushmitha@DESKTOP-607RQ7V:~$ _
```

```
sushmitha@DESKTOP-6O7RQ7V: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
int main()  
{  
for(int i=1;i<=3;i++)  
{  
if(fork()==0)  
{  
printf("Child %d PID = %d\n", i, getpid());  
return 0;  
}  
}  
return 0;  
}
```

5.Linux command using exchlp()


```

sushmitha@DESKTOP-607RQ7V:~$ nano linux.c
sushmitha@DESKTOP-607RQ7V:~$ gcc linux.c -o linux
sushmitha@DESKTOP-607RQ7V:~$ ./linux
Before exec()
total 360
-rwxr-xr-x 1 sushmitha sushmitha 16128 Aug 13 05:43 Multichild
-rw-r--r-- 1 sushmitha sushmitha 255 Aug 13 05:43 Multichild.c
-rwxr-xr-x 1 sushmitha sushmitha 16312 Aug 13 05:17 Week1_1
-rw-r--r-- 1 sushmitha sushmitha 443 Aug 13 04:52 Week1_1.c
-rwxr-xr-x 1 sushmitha sushmitha 16168 Aug 13 05:24 Week1_2
-rw-r--r-- 1 sushmitha sushmitha 443 Aug 13 05:24 Week1_2.c
-rwxr-xr-x 1 sushmitha sushmitha 16088 Aug 12 06:51 childusingfork
-rw-r--r-- 1 sushmitha sushmitha 273 Aug 12 06:50 childusingfork.c
-rw-r--r-- 1 sushmitha sushmitha 98 Aug 3 07:42 doublefor.c
-rwxr-xr-x 1 sushmitha sushmitha 16000 Aug 3 07:47 doublefork
-rw-r--r-- 1 sushmitha sushmitha 97 Aug 3 07:47 doublefork.c
-rwxr-xr-x 1 sushmitha sushmitha 15992 Aug 3 07:39 fork
-rw-r--r-- 1 sushmitha sushmitha 89 Aug 3 07:38 fork.c
-rwxr-xr-x 1 sushmitha sushmitha 16208 Aug 13 05:36 fork1
-rw-r--r-- 1 sushmitha sushmitha 310 Aug 13 05:36 fork1.c
-rw-r--r-- 1 sushmitha sushmitha 13 Aug 13 05:16 input.txt
-rwxr-xr-x 1 sushmitha sushmitha 15992 Aug 13 05:56 linux
-rw-r--r-- 1 sushmitha sushmitha 157 Aug 13 05:55 linux.c
-rwxr-xr-x 1 sushmitha sushmitha 16048 Aug 12 07:06 multiplechid
-rw-r--r-- 1 sushmitha sushmitha 162 Aug 12 07:05 multiplechid.c
-rwxr-xr-x 1 sushmitha sushmitha 16000 Aug 12 06:37 myprocessid
-rw-r--r-- 1 sushmitha sushmitha 112 Aug 12 06:35 myprocessid.c
-rw-r--r-- 1 sushmitha sushmitha 13 Aug 13 05:25 output.txt
-rwxr-xr-x 1 sushmitha sushmitha 16040 Aug 3 08:03 parent
-rw-r--r-- 1 sushmitha sushmitha 212 Aug 3 08:03 parent.c
-rwxr-xr-x 1 sushmitha sushmitha 16000 Aug 1 05:45 parentchild
-rw-r--r-- 1 sushmitha sushmitha 163 Aug 1 06:06 parentchild.c
-rwxr-xr-x 1 sushmitha sushmitha 16088 Aug 12 07:00 parentwaitsforchild
-rw-r--r-- 1 sushmitha sushmitha 260 Aug 12 07:00 parentwaitsforchild.c
-rw----- 1 sushmitha sushmitha 1 Jul 27 06:00 process.c.save
-rwxr-xr-x 1 sushmitha sushmitha 16000 Aug 3 07:35 processid
-rw-r--r-- 1 sushmitha sushmitha 98 Aug 3 07:35 processid.c
drwxr-xr-x 2 sushmitha sushmitha 4096 Jul 29 08:06 test
-rwxr-xr-x 1 sushmitha sushmitha 16056 Aug 3 07:08 username
-rw-r--r-- 1 sushmitha sushmitha 134 Aug 3 07:08 username.c
-rwxr-xr-x 1 sushmitha sushmitha 16032 Aug 3 07:51 wait
-rw-r--r-- 1 sushmitha sushmitha 209 Aug 1 06:01 wait.c
-rwxr-xr-x 1 sushmitha sushmitha 15952 Aug 3 07:12 welcome
-rw-r--r-- 1 sushmitha sushmitha 84 Aug 3 07:11 welcome.c
sushmitha@DESKTOP-607RQ7V:~$

```

```
sushmitha@DESKTOP-607RQ7V: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
int main()  
{  
printf("Before exec()\n");  
execlp("ls","ls","-l",NULL);  
printf("This will not execute.\n");  
return 0;  
}
```

6.Parent and Child executing concurrently

```
sushmitha@DESKTOP-607RQ7V:~$ nano concurrently.c  
sushmitha@DESKTOP-607RQ7V:~$ gcc concurrently.c -o concurrently  
sushmitha@DESKTOP-607RQ7V:~$ ./concurrently  
PID: 1391 Count: 1  
PID: 1392 Count: 1  
PID: 1392 Count: 2  
PID: 1391 Count: 2  
PID: 1392 Count: 3  
PID: 1391 Count: 3  
PID: 1391 Count: 4  
PID: 1392 Count: 4  
PID: 1392 Count: 5  
PID: 1391 Count: 5  
sushmitha@DESKTOP-607RQ7V:~$
```

```
sushmitha@DESKTOP-607RQ7V: ~  
GNU nano 8.7.1 concurrentl  
#include <stdio.h>  
#include <unistd.h>  
int main()  
{  
fork();  
for(int i=1;i<=5;i++)  
{  
printf("PID: %d Count: %d\n", getpid(), i);  
sleep(1);  
}  
return 0;  
}
```

7.zombie process

```
sushmitha@DESKTOP-607RQ7V:~$ nano zombie.c  
sushmitha@DESKTOP-607RQ7V:~$ gcc zombie.c -o zombie  
sushmitha@DESKTOP-607RQ7V:~$ ./zombie  
Child Exiting  
Parent Finished
```

```
sushmitha@DESKTOP-607RQ7V: ~
GNU nano 8.7.1 zombie.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child Exiting\n");
        return 0;
    }
    else
    {
        sleep(15);
        printf("Parent Finished\n");
    }
    return 0;
}
```

8.orphanprocess

```
sushmitha@DESKTOP-607RQ7V:~$ nano orphanprocess.c
sushmitha@DESKTOP-607RQ7V:~$ gcc orphanprocess.c -o orphanprocess
sushmitha@DESKTOP-607RQ7V:~$ ./orphanprocess
Parent Exiting
sushmitha@DESKTOP-607RQ7V:~$ Child PID : 1513
New Parent PID : 406

sushmitha@DESKTOP-607RQ7V:~$
```

```
sushmitha@DESKTOP-607RQ7V:~$ nano orphanprocess.c
sushmitha@DESKTOP-607RQ7V:~$ gcc orphanprocess.c -o orphanprocess
sushmitha@DESKTOP-607RQ7V:~$ ./orphanprocess
Parent Exiting
sushmitha@DESKTOP-607RQ7V:~$ Child PID : 1513
New Parent PID : 406

sushmitha@DESKTOP-607RQ7V:~$
```